

Continuous semantic change monitoring of land cover with Landsat time series

1st Haixu He

China University of Geosciences
School of computer science
Wuhan, China
20161001925@cug.edu.cn

2nd Jining Yan

China University of Geosciences
School of computer science
Wuhan, China
yanjn@cug.edu.cn

Abstract—Rapid urbanization has led to frequent land cover changes, making continuous monitoring essential for capturing land use dynamics and enabling quick responses. It is crucial to detect both the dates and types of these changes, which current change detection models struggle to achieve simultaneously. In this study, we propose a supervised approach called Temporal Semantic Recognition Change Detection (TSRCD), using a Transformer-based sliding detector to learn mappings from spectral data to land cover changes over time. During monitoring, the model utilizes previously detected land cover types and additional observations as dual prompts to iteratively predict the next time point's land cover type. We validated our approach using all clear Landsat data from the Wuhan urban agglomeration between 2008 and 2022. Experiments show that increasing the peek window size improves change detection accuracy, stabilizing at around ten, with an approximate 8% improvement in F1 score compared to CCDC. TSRCD not only tracks the dates of land cover changes but also accurately identifies change types, achieving 92% accuracy. In summary, TSRCD provides a novel and comprehensive solution for monitoring land cover changes using long-term series remote sensing data.

Index Terms—TSRCD, Change monitoring, Landsat time series, Transformer, Deep learning

I. INTRODUCTION

With contemporary society's increasing focus on climate change, environmental protection, and sustainable global resource utilization, land change science has emerged as a crucial scientific field to address these complex challenges [1]. In recent years, due to the increasing demand from scholars for land change analysis, we have observed significant shifts in research trends.

1) Stronger Timeliness: The paradigm shift from change detection at two time points to time series change detection, and further to continuous time series change monitoring [2], wherein continuous time series change monitoring represents an online detection method [3], allowing models to adapt and make judgments as new observations are added to the sequence [4]. Representative methods include continuous change detection (CCD) and its derivative algorithms [5].

2) More Comprehensive Change Information: Change detection has evolved from simply comparing images taken at different times in the same area and displaying changes and their locations, to encompassing various aspects such as change location, time, type, degree, and agent [1]. The

primary challenge in current change detection methods is how to comprehensively obtain this information. Although there has been research in these areas, some argue that the focus remains predominantly on the technical aspects of change detection (where and when), with insufficient attention to change types [1]. The change types referred to in this context specifically mean changes in land cover types. In change monitoring, when land cover changes are detected, newly observed satellite data may lack sufficient temporal evolution features, complicating classification due to phenological factors.

3) More advanced and complex models: In recent years, deep learning methods have been widely adopted by scholars for time series classification tasks due to their excellent capability to automatically learn features [6]. However, despite land cover change detection also belonging to time series analysis tasks [8], deep learning methods have still not been widely applied. We believe that the difficulty in matching the performance of advanced machine learning change detection methods like CCDC or BFAST [4], [5] is mainly due to the lack of an appropriate framework to address challenges such as uneven temporal distribution of observations (data issue), difficulty in setting change thresholds (method issue), and the requirement for continuous monitoring (application issue).

The shift in these research trends is evident, leaving ample research space. This study will focus on using deep learning to address continuous land cover change monitoring tasks. Building upon the aforementioned shift in research trends, we will contribute our solution based on the following two questions.

- 1) How to achieve continuous change monitoring while obtaining change type information?
- 2) If utilizing the deep learning model to solve the change monitoring task, what will the model architecture look like, and what will be the inputs and outputs?

For the first question, it is essential to emphasize how to achieve accurate intra-annual land cover classification tasks while disregarding phenological information [9]. Previous works, such as He et al.'s research [10], densely annotated all timestamps in the time series, providing a feasible solution for high-frequency land cover classification. However, these methods are often offline and less suitable for change monitoring. In

this regard, algorithms like CCD series offer valuable insights into change monitoring [5]. These algorithms utilize pre-fitted models to predict the next observation, and significant prediction errors indicate land cover changes. Building upon this idea, we can upgrade the task from predicting the spectral value of the next observation to predicting the land cover of the next observation, thereby better addressing the first issue.

For the second question, accurately predicting land cover at the next observation requires considering not only the spectral features of the current observation but also the spectral values or land cover in the interval before and after it. These key features are essential for helping the model disregard phenological factors and achieve precise land cover classification at a single time point [11]. Our task flow can be viewed as a remote sensing satellite time series translation task, where the goal is to translate the land cover type corresponding to each time point or different types of change information over time. This process is expected to use an autoregressive approach, considering the aforementioned features for each land cover type prediction. Deep learning models based on the Transformer architecture [12], are effective for such translation tasks and can serve as the primary model backbone for this problem.

In this study, we propose the Temporal Semantic Recognition Change Detection (TSRCD) method for continuous change monitoring of time series data with the ability to detect change types. In the sample generation phase, we improved a sample synthesis method aimed at providing fine-grained land cover labels at each time point for deep models. At the algorithmic level, we introduce a change detector based on the Transformer architecture, capable of effectively monitoring semantic land cover changes in remote sensing time series.

II. METHOD AND MATERIAL

A. Transformer-based change detector

In the TSRCD, a model based on the Transformer architecture serves as a change detector applied to remote sensing time series for monitoring land cover change. It is worth noting that, in theory, time-series processing schemes based on recurrent neural networks can fully replace Transformer. The overall monitoring process of TSRCD is illustrated in Fig 1. Firstly, we introduce the network structure and input-output of the Transformer, and then we elaborate on the monitoring process.

1) *Embedding layer*: For the time series sample to be detected, denoted as $X \in \mathbb{R}^{L \times C}$, where L represents the input sequence length, and C denotes the number of available spectral bands. In order to fully utilize the information in the sequence, we designed three embedding layers for the model as inputs:

- 1) *Token embedding*: To represent the sequence, we employ one-dimensional convolution to map the sequence to a feature space of dimension d_{model} .
- 2) *Positional embedding*: Since the self-attention mechanism operates on input embeddings simultaneously and identically, the sequence order is ignored. To utilize

sequence information and preserve its order, a common solution is to append additional positional encodings at the input end. A typical choice is to use sine and cosine functions with different frequencies.

- 3) *Temporal embedding*: Temporal embedding is designed to capture the order and intervals of elements in a sequence. We introduce timestamp information, encoding the day of year (DOY) for each timestamp, to handle uneven timestamps. This embedding layer improves the model's understanding of temporal correlations, enhancing its capability to model uneven time series.

Finally, by summing up the token, position, and temporal embedding together, we suggest concatenating the tokens for begin of sequence (BOS) and end of sequence (EOS) respectively at the beginning and end of the sequence. This is to facilitate controlling the model's output length in tasks where the input and output sequences are of unequal lengths. Ultimately, the vector $X_{in} \in \mathbb{R}^{(L+2) \times d_{model}}$ is inputted into the Transformer model. The same embedding approach is applied to the input of the decoder.

2) *Transformer structure*: We explore using the native Transformer architecture for change monitoring without structural modifications [13]. The Transformer comprises an encoder and decoder, each with N layers. The encoder uses self-attention to extract features, focusing on critical moments in the time series. The decoder works with the encoder to classify the land cover type of the next observation.

In the encoder, the main task is to map the input sequence to hidden representations that capture temporal relationships. In the decoder, which receives a BOS token indicating the start of the sequence. The decoder's hidden state, combined with the compressed representation of the original sequence from the encoder, is used to predict the land cover type at the first time point. This prediction is then embedded after the BOS token and iteratively input into the decoder using vector autoregression until the EOS token is received. At each time step, the model adjusts its focus based on previous predictions and hidden representations, capturing subtle patterns of land cover change more accurately. This process completes the identification of land cover changes, and by analyzing the time points of these changes, the specific dates are determined.

B. TSRCD monitoring process

Continuous change monitoring involves intensive land cover monitoring using all clear Landsat observations. Inspired by CCDC [14], we use a moving window (mw) for continuous monitoring to mitigate the impact of anomalies or earlier changes. We also introduced a hyperparameter, peek size (ps), to control the peek window's size, allowing the model to make additional observations before classifying the current observation's land cover type. For example, with $mw = 16$ and $ps = 5$ (see Fig 2), the moving window covers 16 observations. When classifying the 12th observation, the model uses the classified results of the previous 11 observations and considers the 5 unclassified observations in the peek window to determine the land cover type. In essence, our strategy is similar to

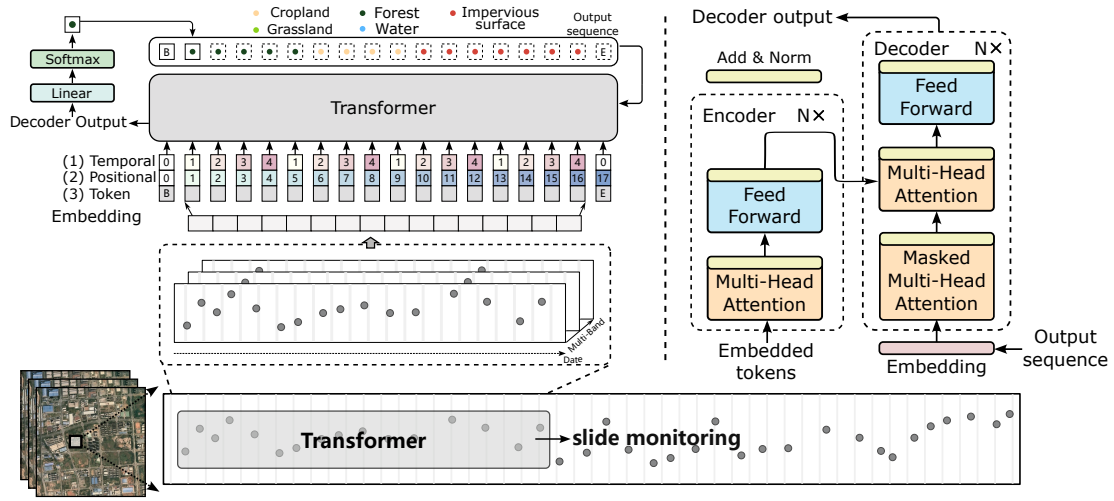


Fig. 1. The TSRCD model structure diagram proposed for this study.

CCDC, but instead of predicting the spectral value of the next observation, we predict the land cover type.

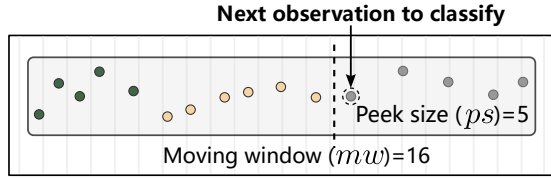


Fig. 2. Schematic diagram of how the hyperparameters mw and ps control the model monitoring process.

C. Experimental configuration

All experiments were conducted using the same Transformer architecture, with the number of stacked layers N in both the encoder and decoder set to 6. The dimensionality of input and output vectors d_{model} was set to 256, the dimensionality of the feed-forward fully connected layers d_{ff} was set to 512, and the number of attention heads h was set to 8. We choose the cross-entropy loss function commonly used in semantic segmentation tasks as the optimization guide of the model. In this study, it is used to calculate the multi-classification loss of land cover in different months. The cross-entropy loss function is defined as:

$$Loss = -\frac{1}{L} \frac{1}{N} \sum_i \sum_{c=1}^M y_{ic} \log(p_{ic}) \quad (1)$$

where L represents the number of observations for each sample, N and M represent the number of samples and categories respectively, y_{ic} is a sign function, if the true category of sample i is equal to c , it takes 1, otherwise it takes 0, p_{ic} represents the predicted probability that sample i belongs to category c .

D. Material

We selected the Wuhan urban agglomeration (WUA) for our study (Fig 3), where urbanization has significantly expanded

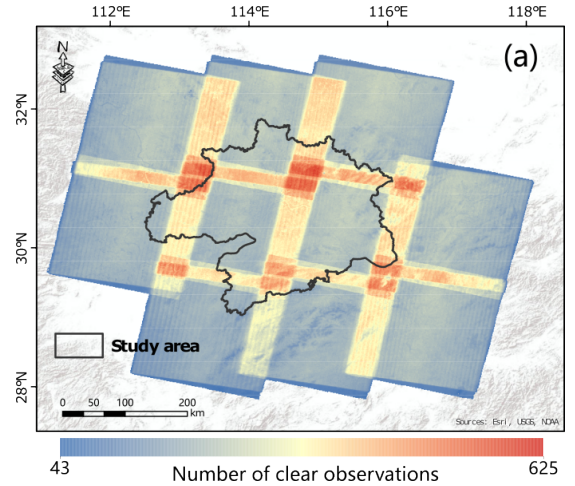


Fig. 3. Landsat data information covering Wuhan urban agglomeration from 2008 to 2022.

construction land from 250.66 km^2 in 2000 to 947.65 km^2 in 2020. Using Landsat TM/ETM+/OLI/OLI2 data (2008-2022) with less than 80% cloud cover from the Google Earth Engine, we analyzed up to 625 clear observations. The dataset included red, green, blue, NIR, SWIR1, and SWIR2 bands. We classified the area into five categories: cropland, forest, impervious surface, grassland, and water.

III. EXPERIMENT

A. Accuracy assessment

1) *Change detection assessment:* The accuracy evaluation considered commission error, omission error, and F1 score. Commission error occurs when the algorithm detects non-existent disturbances, while omission error is when actual disturbances are missed. The F1 score balances these errors. Each disturbance event in the reference plots was evaluated for spatial and temporal accuracy. The study covered 15 years

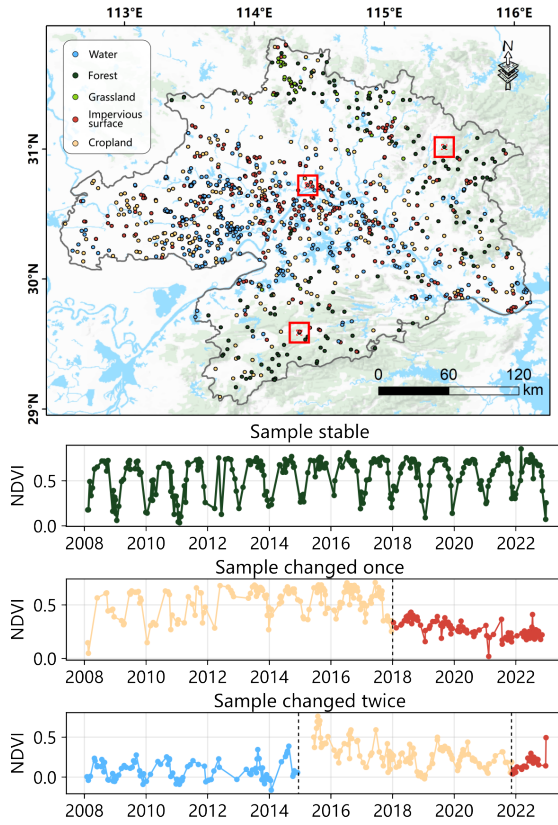


Fig. 4. The spatial distribution of the test sets for all experiments in this study, along with the data and labels of three typical samples (presented by NDVI).

(2008-2022) and assessed 15,000 cases using 1,000 Landsat time series samples (500 stable and 500 change samples).

2) *Classification assessment*: For each test sample, we calculated overall accuracy (OA) for land cover classification and introduced a metric called Change Type Detection Accuracy. This metric measures the proportion of correctly identified change types in the test set. Even stable samples are considered to have a “stable” change type. As shown in Fig 4, change type detection is accurate only if the TSRCD model identifies the types correctly, such as forest, cropland to impervious surface, and water to cropland to impervious surface. Note that TSRCD requires additional steps to record the sequence of land cover occurrences to determine the change type.

B. Configuration

The experiment was conducted using PyTorch version 2.1.0, with a batch size set to 64. The NoamOpt optimizer from the original Transformer model was employed, with reference to the GitHub code available at <https://github.com/HaixuHe/Continuous-change-detection-Of-Land-cover-using-Transformer>.

IV. RESULTS

In this section, we will demonstrate and analyze the performance of TSRCD in three different change detection scenarios. We will combine corresponding experimental analyses to study

the spatiotemporal development patterns of the research area from 2008 to 2022, spanning 15 years.

A. Change detection and classification accuracy assessment

Fig 5 (a) presents the variation of the F1 score under different ps conditions for various mw . As ps increases, the improvement in the model’s change detection capability is evident. However, a performance bottleneck is reached when ps equals ten. Additionally, we found that the model with mw equal to 50 achieves the best results compared to other models. At this point, the window approximately covers a three-year period (with an average of 17 clear observations per year). Larger time windows do not yield better performance. We believe this may be because larger windows encompass too many time steps, making it more challenging for the model to pinpoint the specific time point corresponding to each land cover type output. Fig 5 (b) presents the performance under different ps conditions when mw equals 50. When ps is small, the detector has fewer land cover types to infer. For example, when ps equals 1, the model, guided by the previous $mw-1$ inference results, tends to prioritize the consistency of land cover types, potentially overlooking actual land cover changes. In this case, Omission error is higher than Commission error. As ps increases, the model has more reference information, leading to a significant reduction in change detection errors. We focus on the performance when ps is less than 10, as in practical applications, a slight performance decrease due to fewer observations is an acceptable trade-off. This also facilitates comparison with the CCDC model. We selected the pyCCD version used in the LCMAP project [15], using default parameters except for PEEK_SIZE. We can see that when ps is less than 8, the accuracy of the TSRCD model is comparable to that of CCDC. As ps increases, TSRCD achieves higher accuracy, achieving an F1 score improvement of about 8% over CCDC, demonstrating its utility. In terms of classification accuracy evaluation, OA remains stable at 93.9, while change type accuracy stabilizes at 92.

Fig 6 shows TSRCD monitoring land cover change in Landsat time series. Panels (a) to (e) illustrate detector states at different observations. The curves indicate probabilities of land cover types from the decoder’s final classification head post-Softmax. Red and green dashed circles highlight the current and look-ahead window observations. In state (a), the detector classifies the observation as cropland with nearly 100% certainty. As new observations are added, state (b) shows a significant decrease in cropland probability, and by state (c), water is detected as the dominant type, marking a land cover change. State (d) confirms water classification with near 100% certainty. By the end of 2022, state (e) shows no further changes detected. TSRCD continues monitoring with new observations, while CCDC requires at least MEOW_SIZE (12 by default) clear observations to initialize the harmonic model and derive stable fitting coefficients and RMSE for land cover classification [15].

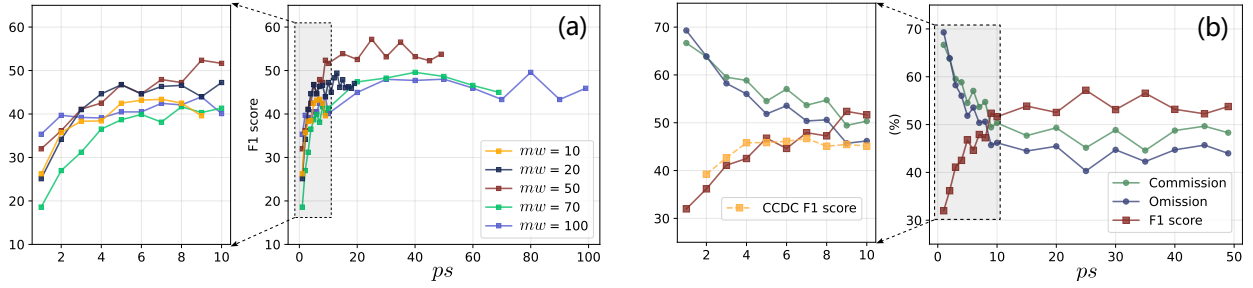


Fig. 5. The variation in change accuracy of the TSRCD model under different moving windows (mw) and peek size (ps) hyperparameters.

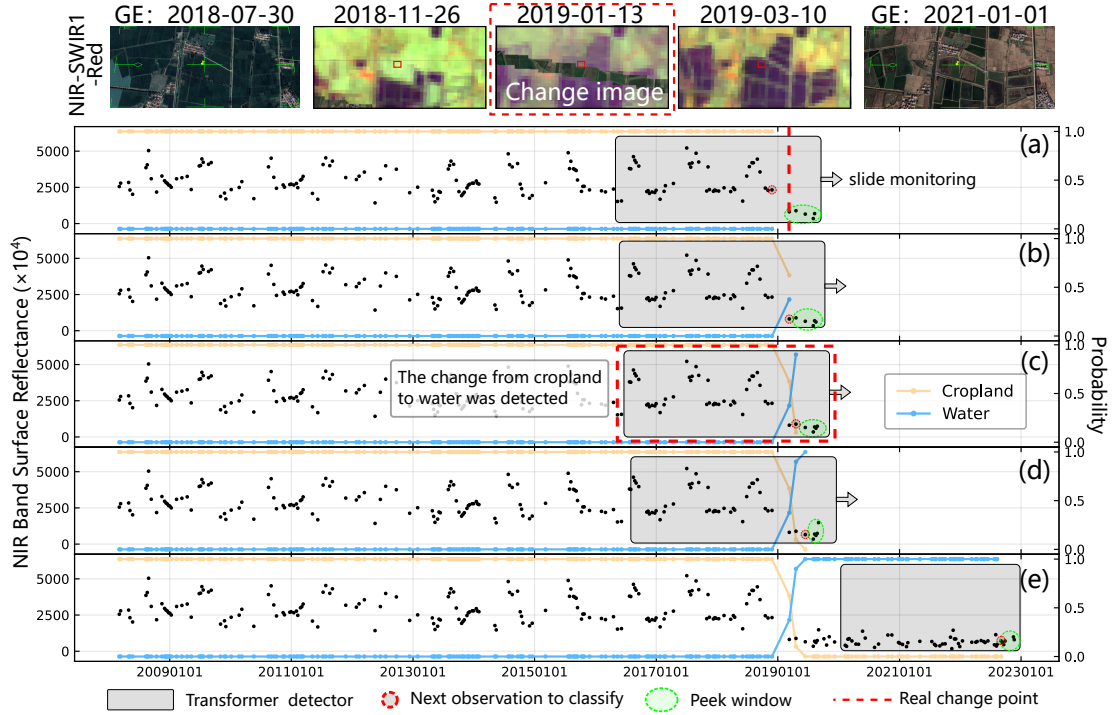


Fig. 6. A case of using TSRCD ($mw=50$, $ps=6$) to monitor land cover change in Landsat time series. GE: Google earth.

B. Land change map with TSRCD

We used the TSRCD model with $mw=50$ and $ps=6$ to infer the change mapping of the entire WUA (Fig 7). Fig 7(a) shows the change map of the WUA, while Fig 7(b-d) display some magnified regions with more detailed changes and visualize the cumulative change maps for 2012, 2017, and 2022. Representative changes were selected within each region, with Google Earth high-resolution imagery added as a reference. In Fig 7(b), the conversion from dryland to paddy fields is shown (although both dryland and paddy fields are classified as cropland from the perspective of land resource utilization [1], here dryland specifically refers to cropland, while paddy fields are classified as water). Fig 7(c) depicts both the decrease and recovery of forest with the upper example demonstrating a decrease in forest land and the lower one showing a case of forest land recovery. Fig 7(d) illustrates the addition of impervious surface. It can be observed that TSRCD performs well in accurately detecting change date.

V. CONCLUSIONS

In this study, we propose a novel deep learning model for continuous monitoring land cover changes in irregular time series called TSRCD. The main idea is to implement a sliding detector based on the Transformer architecture on the time series. This goal is largely achieved by introducing land cover annotations for the detector at each time point, helping the model learn the dense mapping from spectral to land cover over time. The detector outputs the land cover type for the next time point in an autoregressive manner during sliding monitoring, aiming to achieve continuous monitoring of land cover changes. As far as we know, this approach represents the first application of using deep learning models to solve the task of time series land cover change monitoring.

ACKNOWLEDGMENT

This research has been supported by the Open Research Program of the International Research Center of

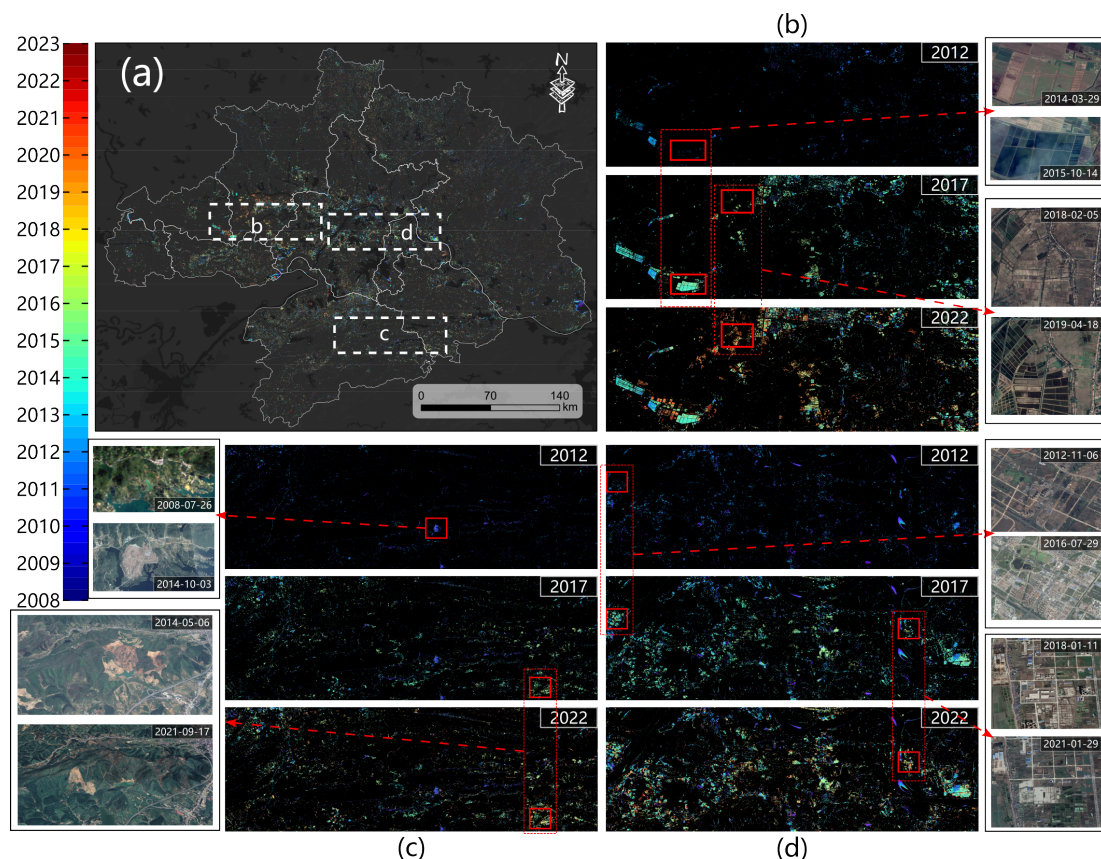


Fig. 7. Land change map of Wuhan urban agglomeration from 2008 to 2022 based on TSRC model inference and display of some key areas.

Big Data for Sustainable Development Goals under Grant CBAS2023ORP03 and National Natural Science Foundation of China (41925007).

REFERENCES

- [1] Z. Zhu, S. Qiu, and S. Ye, "Remote sensing of land change: A multifaceted perspective," *Remote Sensing of Environment*, vol. 282, p. 113266, 2022.
- [2] C. E. Woodcock, T. R. Loveland, M. Herold, and M. E. Bauer, "Transitioning from change detection to monitoring with remote sensing: A paradigm shift," *Remote Sensing of Environment*, vol. 238, p. 111558, 2020.
- [3] Z. Zhu, "Change detection using landsat time series: A review of frequencies, preprocessing, algorithms, and applications," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 130, pp. 370–384, 2017.
- [4] J. Verbesselt, A. Zeileis, and M. Herold, "Near real-time disturbance detection using satellite image time series," *Remote Sensing of Environment*, vol. 123, pp. 98–108, 2012.
- [5] Z. Zhu and C. E. Woodcock, "Continuous change detection and classification of land cover using all available landsat data," *Remote sensing of Environment*, vol. 144, pp. 152–171, 2014.
- [6] N. Kussul, M. Lavreniuk, S. Skakun, and A. Shelestov, "Deep learning classification of land cover and crop types using remote sensing data," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 5, pp. 778–782, 2017.
- [7] L. Zhong, L. Hu, and H. Zhou, "Deep learning based multi-temporal crop classification," *Remote sensing of environment*, vol. 221, pp. 430–443, 2019.
- [8] A. Mullissa, J. Reiche, and M. Herold, "Deep learning and automatic reference label harvesting for sentinel-1 sar-based rapid tropical dry forest disturbance mapping," *Remote Sensing of Environment*, vol. 298, p. 113799, 2023.
- [9] C. Karakizi, I. Tsiotas, Z. Kandalakis, A. Vaiopoulos, and K. Karantzalos, "Assessing the contribution of spectral and temporal features for annual land cover and crop type mapping," *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, vol. 43, pp. 1555–1562, 2020.
- [10] H. He, J. Yan, D. Liang, Z. Sun, J. Li, and L. Wang, "Time-series land cover change detection using deep learning-based temporal semantic segmentation," *Remote Sensing of Environment*, vol. 305, p. 114101, 2024.
- [11] V. F. Rodriguez-Galiano, B. Ghimire, J. Rogan, M. Chica-Olmo, and J. P. Rigol-Sanchez, "An assessment of the effectiveness of a random forest classifier for land-cover classification," *ISPRS journal of photogrammetry and remote sensing*, vol. 67, pp. 93–104, 2012.
- [12] S. P. Singh, A. Kumar, H. Darbari, L. Singh, A. Rastogi, and S. Jain, "Machine translation using deep learning: An overview," in *2017 international conference on computer, communications and electronics (comptelix)*. IEEE, 2017, pp. 162–167.
- [13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.
- [14] Z. Zhu, J. Zhang, Z. Yang, A. H. Aljaddani, W. B. Cohen, S. Qiu, and C. Zhou, "Continuous monitoring of land disturbance based on landsat time series," *Remote Sensing of Environment*, vol. 238, p. 111116, 2020.
- [15] G. Z. Xian, K. Smith, D. Wellington, J. Horton, Q. Zhou, C. Li, R. Auch, J. F. Brown, Z. Zhu, and R. R. Reker, "Implementation of the ccde algorithm to produce the lcamap collection 1.0 annual land surface change product," *Earth System Science Data*, vol. 14, no. 1, pp. 143–162, 2022.